

Image-based navigation system using skylight polarization for unmanned ground vehicle

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ABSTRACT

With advances in unmanned and autonomous vehicles, camera-based navigation is increasingly being used. A new low-cost navigation solution based on polarization filter array cameras is presented. For this purpose, we developed our own acquisition setup, and adapted an existing image processing to find the relative heading thanks to the skylight polarization. The precision of the method is quantified using a rotary stage, and an Unmanned Ground Vehicle (UGV) from where the estimated heading is compared to a reference given by a GPS/Inertial Navigation System.

Keywords: Polarization Imaging, Navigation, Unmanned Ground Vehicle, Embedded Systems, Computer Vision, Imaging Processing

1. INTRODUCTION

According to the hypothesis of some researchers, the Vikings used the polarization to navigate across the oceans with what they called the "sunstone",^{1,2} which acted as a linear polarizer. Many animals are able to perceive the celestial polarization pattern for navigation,^{3,4} e.g. the North African desert ant.⁵ Camera-based navigation is now widely used, especially in Unmanned Ground Vehicles (UGV). Existing navigation solutions are mainly based on conventional monochrome cameras.^{6,7} A possible extension is to use polarization imaging.^{4,8} With advances in technology and new types of sensors,⁹ it is now easier to capture rapidly the polarization information. The Polarization Filter Arrays (PFA) sensor, with a practical implementation with the SONY IMX250 MZR,⁹ is one of these sensors. It has the advantage to capture the polarization information with a single sensor and in a snapshot way. It is even possible to sense spectro-polarimetric data with this technique,^{10,11} and can benefit for many computer vision applications using both modalities.

The diffusion of the light by the sky atmosphere generates a skylight polarization pattern which mathematically encodes the sun position.¹² It is then possible to estimate the position of the sun using the Rayleigh scattering model.^{13,14}

The aim of this paper is to develop a new navigation solution based on low-cost polarization cameras. To this end, we develop an acquisition system and an image processing algorithm to find the heading of an Unmanned Ground Vehicle (UGV). We evaluate the method by quantifying the orientation errors relatively to an Inertial Measurement Unit (IMU) or a Global Positioning System/Inertial Navigation System (GPS/INS). In Section 2, we describe the acquisition setups to capture images of the sky using PFA cameras. Next, the imaging model and navigation algorithm are defined in Section 3. Finally, the results are presented in Section 4, before the conclusion in Section 5.

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2. ACQUISITION SETUPS

2.1 Imaging & Optics

Capturing sky scenes with dynamic content or under large displacement of the camera needs a fast and efficient device. Thus, our image acquisition is based on a Division of Focal-Plane Polarimeter (DoFP), which allows the capture of different linear polarization states without the need for image registration (due to multiple cameras, for example) ^{.15,16} The advantages of such device are the cost, the compactness, and the snapshot capability. Other polarimetric imaging techniques often require either a rotated polarizer, ¹⁷ or a cumbersome optical configuration. ¹⁸ The DoFP polarimeter uses an array of polarization filters on top of the focal plane array. The periodic pattern consists of four polarization filters oriented at 0°, 45°, 90°, and 135° and constitute a “superpixel” (see Figure 1). The principal issue of these polarimeters is their limited spatial resolution, where a single pixel senses only a particular polarization state. Thus, an additional spatial interpolation algorithm can be needed to recover the full resolution of images.

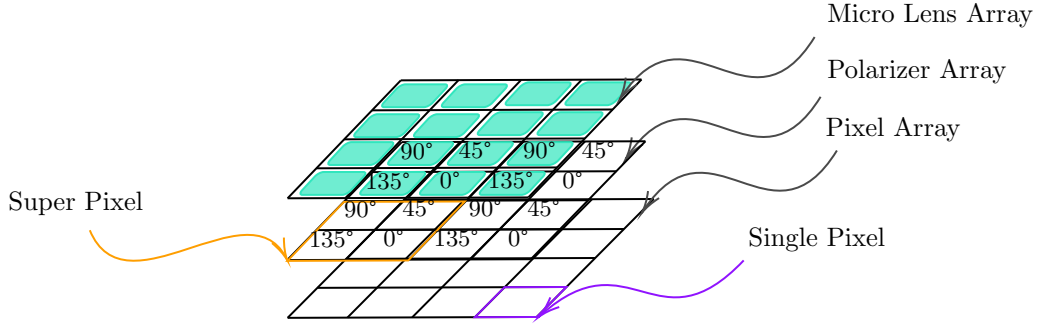


Figure 1: The DoFP polarimeter architecture. Here is the sensor arrangement specific to the Sony IMX250 MZR.

We decided to capture the polarization data in broadband with the following camera:

- Flir Blackfly S (BFS-U3-51S5P-C) with the PFA Sony IMX250 MZR sensor.

Our camera is equipped with a Kowa lens (LM8JC5MC, $f=8\text{mm}/F2.8$), which offers a relatively good compromise between field of view, amount of light, and lens aberration. The impact of the lens parameters is not considered in this work.

2.2 Mechanics

We focused on two main objectives: 1-capture data on both static and dynamic platforms, to sweep a maximum of practical cases, and 2-evaluate robustly the navigation algorithm using redundant sources of reference data.

Two mechanical platforms have been built:

1. **The rotary setup**, for benchmarking the navigation algorithm and evaluating its performance in static configuration (see Figure 2a). It is composed of a mechanical-bearing direct-drive rotary stage (Aerotech ADRS 200, accuracy: $29.1\mu\text{rad}$). The purpose of this setup is to develop the navigation algorithm and to have a feedback on the performances of this algorithm. The camera is mounted on a 90° metal L-shape and the IMU (InertialLabs Kernel 100) is placed on the same reference plane as the rotary stage. This allows to compare the data provided by both the IMU and those provided by the feedback of the ADRS (see Figure 2a). Indeed, the redundancy between the cameras, the IMU, and the rotary bench feedback, allows to quantify the errors with a relatively good precision.

2. **The embedded setup** (UGV), for benchmarking the navigation algorithm under dynamic scenes (see Figure 2b). The embedded setup is made up of a custom UGV. It is equipped with a camera and an IMU. It also embeds a GPS/INS to have reference data to compare with the others sensors. The aim of this system is to have dynamic data to optimize the navigation algorithm according to the different situations encountered.

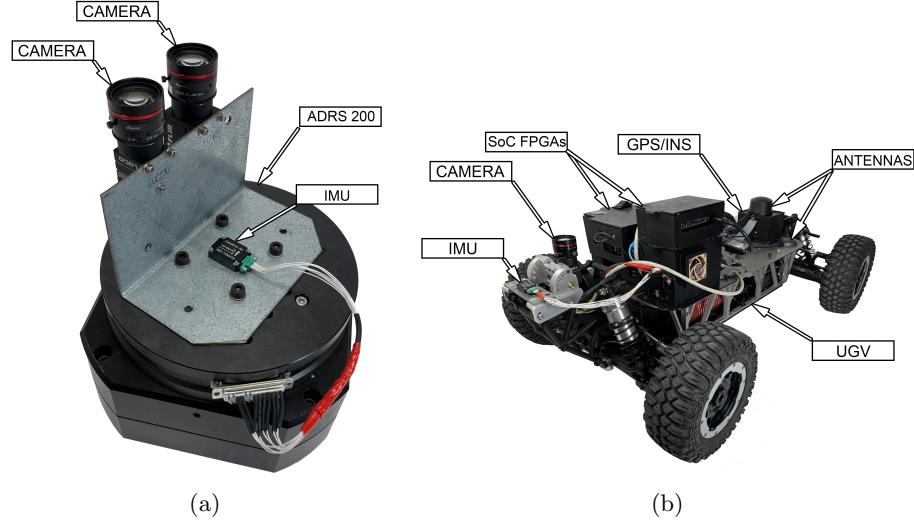


Figure 2: (a) The rotary setup and (b), the embedded setup. For the Figure 2a, the second camera is here for a future work.

2.3 Electronic Boards

The electronic systems allow for the setups to be remotely controllable, under battery or main power supply. To do this, we developed two electronic systems, each of which has a specific function:

1. An embedded computer (Avnet Ultra96-V2) consisting of a system-on-chip (SoC) with an ARM processor and a Xilinx Ultrascale Field-Programmable Gate Array (FPGA) (see Figure 3a). The Avnet board runs an Ubuntu 20.04 Server distribution and allows remote access via a Wi-Fi hub. The function of this system is to configure the cameras and store the sky images in RAW format.
2. A customized trigger daughterboard consisting of a Trenz TE0706 carrier board with a TE0715 SoC Module with Xilinx Zynq-7000 (see Figure 3b). The purpose of this electronic board is to generate the appropriate voltages for the sensors (camera and IMU). It also generates a GNSS trigger signal using a FPGA-based software to synchronize the equipment and therefore the acquisitions. A Digi XBee module provides wireless connectivity to configure the trigger frequency remotely.

In the future, the systems 1 and 2 will be merged into the same SoC for more compactness.

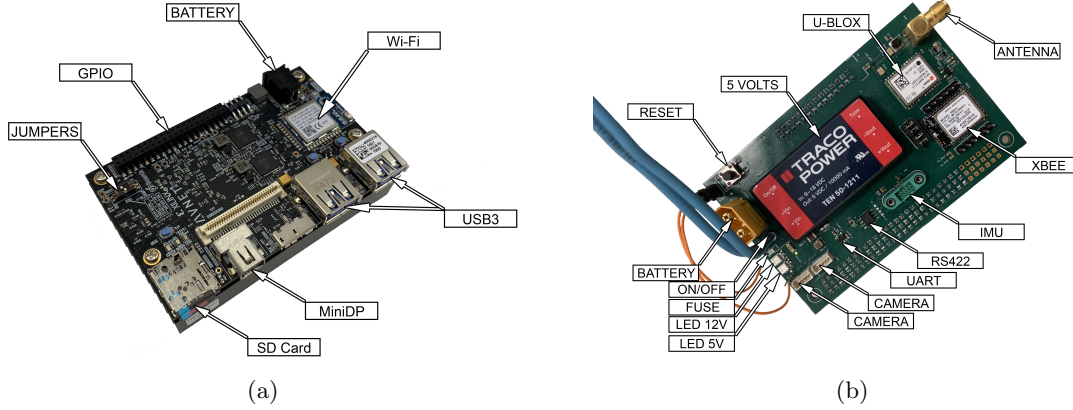


Figure 3: (a) Embedded SoC, (b) Trigger Card.

3. IMAGING & NAVIGATION

3.1 Imaging model

The Stokes vector $\mathbf{S}$ describes the polarization state of the light.¹⁹ It defines a four components vector $\mathbf{S} = [S_0 \ S_1 \ S_2 \ S_3]^t$, with S_0 the total light intensity, S_1 the intensity difference through a 0° and 90° polarizers, S_2 the intensity difference through a 45° and 135° polarizers, and S_3 referring to left or right handedness of the polarized light. The formalism includes whether the light is linearly or circularly polarized, and whether it is fully polarized, partially polarized, or unpolarized. In case of capturing only the linear polarization states (our case, circular polarization being rarely present in nature), S_3 is fixed to zero.

A polarimeter is a measurement device used to estimate the Stokes vector. The output of such device is a vector $\mathbf{I}$ containing M measured intensities, captured with different polarization filter orientations θ . Assuming no additive noise, the measurement is related to the Stokes parameters by a linear model as follows:

$$\mathbf{I} = \mathbf{A} \mathbf{S}, \quad (1)$$

where $\mathbf{A}$ is the analyzer matrix, also called the data reduction matrix. A calibration procedure is often employed to determine $\mathbf{A}$.²⁰ The PFA camera used in this work (and described in Section 2.1) has $M = 4$ linear polarizers oriented at $\theta \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ\}$. We assume that this polarimeter is ideal,²¹ i.e. the polarizing elements have high extinction ratio and low angular errors, so that the calibration of $\mathbf{A}$ can be omitted ($\mathbf{A} = \mathbf{A}_{ideal}$). From the camera, we get four measurements, which leads to:

$$\mathbf{I} = \begin{bmatrix} I_0 \\ I_{45} \\ I_{90} \\ I_{135} \end{bmatrix} = \mathbf{A}_{ideal} \mathbf{S} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix}. \quad (2)$$

Then, we estimate the Stokes vector $\hat{\mathbf{S}}$ as follows:

$$\hat{\mathbf{S}} = \mathbf{A}_{ideal}^+ \mathbf{I}, \quad (3)$$

where the subscript $+$ means the pseudo-inverse estimator.

As seen in Figure 1, the 2D sensor arrangement consists in a periodic pattern called “superpixel”, where each pixel senses only one polarized intensity. Thus, we subsample the raw image of size n by m to organize all the superpixels in a $\frac{n}{2} \times \frac{m}{2} \times 4$ matrix. Finally, we apply the Equation (3) for each superpixel. Another solution would be to use a dedicated demosaicking algorithm²² to get an estimate of $\mathbf{I}$ at each pixel position (giving an

image of size $n \times m \times 4$). Nevertheless, we do not use demosaicking algorithm because 1-the sky data is assumed to be highly correlated in the lowpass frequencies, and 2-this process can introduce undesirable additional noise.

The Angle of Linear Polarization (AoLP) indicates the angle between the reference axis of the sensors and the polarization direction of the light. It is defined as follows:

$$\phi = \frac{1}{2} \arctan \frac{S_2}{S_1} . \quad (4)$$

The Degree of Linear Polarization (DoLP) describes the part of an electromagnetic wave that is linearly polarized. Its value is in the range $[0, 1]$, computed as follows:

$$\rho = \frac{\sqrt{S_1^2 + S_2^2}}{S_0} . \quad (5)$$

3.2 Rayleigh Model

The sun's rays passing through the small particles that make up the Earth's atmosphere are polarized according to the Rayleigh scattering model.^{13,14} The Rayleigh scattering is a special case of the Mie theory,²³ only valid for molecules and particles of dimensions much smaller than the wavelength. It appears that the skylight polarization depends on the position of the sun and on the direction of the observation.

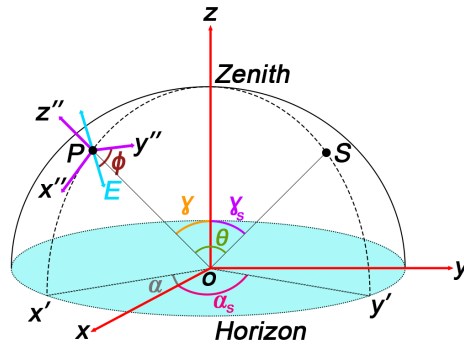


Figure 4: Skylight polarization: E-vector direction in a local frame with γ : zenith angle, α : azimuth angle, ϕ : polarization angle, and θ : scattering angle. S belongs to the Sun.

Figure 4 describes the geometrical parameters involved in the Rayleigh scattering model. O represents the observation position. P is the direction of observation in the celestial sphere with γ the zenith angle and α the azimuth angle. S is the direction of the sun in the celestial sphere, with γ_s the zenith angle and α_s the azimuth angle. ϕ is the polarization angle of the scattered light that can be measured with a polarized camera. θ is the scattering angle. $OXYZ$ is the camera frame. $OX'Y'Z'$ is the meridian frame. $OX''Y''Z''$ is the incident ray frame. According to the Rayleigh scattering model, the scattered light has a vector $\vec{E}$ oriented perpendicular to the scattering plane (OPS). All polarization directions $\vec{E}$ are perpendicular to the sun vector $\vec{OS}$.

According to Figure 4 we can define the DoLP and AoLP equations:^{24,25}

$$\rho = \frac{1 - \cos^2(\theta)}{1 + \cos^2(\theta)} , \quad (6)$$

$$\phi = \arctan \left(\frac{\sin(\alpha - \alpha_s) \sin(\gamma_s)}{\cos(\gamma_s) \sin(\gamma) - \sin(\gamma_s) \cos(\gamma) \cos(\alpha - \alpha_s)} \right) . \quad (7)$$

The AoLP is equal to 0° or 180° when $\alpha = \alpha_s$ and $\alpha = \alpha_s + 180^\circ$. The DoLP is equal to 0 for $\gamma = \gamma_s$ or 1 when $\theta = 90^\circ$.

3.3 Navigation algorithm

The navigation algorithm is partly based on Lu *et al.*²⁶ work. The main principle relies on finding a line in the AoLP map, which characterizes the solar meridian (see Equation (7)). First, a threshold-extraction method is used to identify the points in the AoLP map whose values are around $\phi = 0^\circ/180^\circ$ which give the sun direction. Secondly, we estimate a line of best fit in the point cloud, taking into consideration the noise behavior in the image. We will describe these two steps in the following.

3.3.1 Threshold-extraction

First, we subsample the raw camera image by superpixel to construct the four polarization states images $\mathbf{I}_0$, $\mathbf{I}_{45}$, $\mathbf{I}_{90}$, and $\mathbf{I}_{135}$. Then, we compute the AoLP and DoLP maps using Equations (6-8) from the Stokes vectors computed with Equation (3).

It is necessary to perform a change of reference frame in order to pass from the incident ray frame to the meridian frame. The rotation matrix allowing the change of reference frame is as follows:

$$R = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) & 0 \\ -\sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (8)$$

It is applied only on S_1 and S_2 Stokes parameters. We perform an image thresholding to get a binary image $\mathbf{BW}$ which contains the points of the solar meridian as follows:

$$BW(x, y) = \begin{cases} 1 & \text{if } |\phi(x, y)| < T \\ 0 & \text{otherwise} \end{cases}, \quad (9)$$

where x and y are the pixel coordinates in the AoLP map, and T is a threshold for the tolerance around the ϕ value (we choose $T = 0.1^\circ$). An example of a binary image is shown in Figure 5.

We define $\mathbf{P} = \begin{pmatrix} x_1 & x_2 & \dots & x_N \\ y_1 & y_2 & \dots & y_N \end{pmatrix}^t$ which contains the N spatial coordinates $\begin{pmatrix} x_i \\ y_i \end{pmatrix}$ of all pixels labeled with $BW = 1$.

3.3.2 Line fitting

We have substituted the Hough transform from Lu *et al.*²⁶ by our own method to reduce the computational complexity. In theory, due to Rayleigh scattering, the direction of the sun should be a line in the camera plane. We define the general model of this line such as:

$$2a_1x - 2a_2y + b = 0, \quad (10)$$

where the coefficients $(a_1, a_2, b) \in \mathbb{R}^3$ are unknowns. We arrange these coefficients in a vector $\mathbf{K} = \begin{pmatrix} a_1 \\ a_2 \\ b \end{pmatrix}^t$ for compactness.

From the set of point coordinates $\mathbf{P}$, we perform the Maximum Likelihood Estimation (MLE). We assume that the noise that perturbs the intensity measurements is additive and Gaussian,²⁷ so that the AoLP values tend to follow a Gaussian distribution. Thus, we define a Gaussian penalty function,²⁸ where the density of probability is defined as:

$$pdf(x, y) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2} \left(\frac{2a_1x - 2a_2y + b}{\sigma}\right)^2\right). \quad (11)$$

To find the values of a_1, a_2 , and b in Equation (10), we apply the log likelihood function L on the set of coordinates such as:

$$\log(L(\mathbf{P}, \mathbf{K})) = -\frac{N}{2} \log(2\pi) - N \log(\sigma) - \frac{1}{2} \sum_{i=1}^N \left(\frac{2a_1 x_i - 2a_2 y_i + b}{\sigma} \right)^2. \quad (12)$$

Then, the log likelihood function is maximized by solving the following equation:

$$\frac{\partial \log(L(\mathbf{P}, \mathbf{K}))}{\partial \mathbf{K}} = 0, \quad (13)$$

$$\Leftrightarrow \begin{bmatrix} N & 2 \sum_{i=1}^N x_i & -2 \sum_{i=1}^N y_i \\ 2 \sum_{i=1}^N x_i & 4 \sum_{i=1}^N x_i^2 & -4 \sum_{i=1}^N x_i y_i \\ -2 \sum_{i=1}^N y_i & -4 \sum_{i=1}^N x_i y_i & 4 \sum_{i=1}^N y_i^2 \end{bmatrix} \begin{bmatrix} b \\ a_1 \\ a_2 \end{bmatrix} = 0. \quad (14)$$

Solving Equation (14) gives the coefficients of the line and thus the direction of the sun. The DoLP could then be used to solve the 180° indeterminacy on the direction of the sun.

Figure 5 shows the complete processing pipeline for the navigation algorithm: we first acquire a raw image of the sky, compute the polarization parameters, estimate the heading, and display the solar meridian in a polar plot.

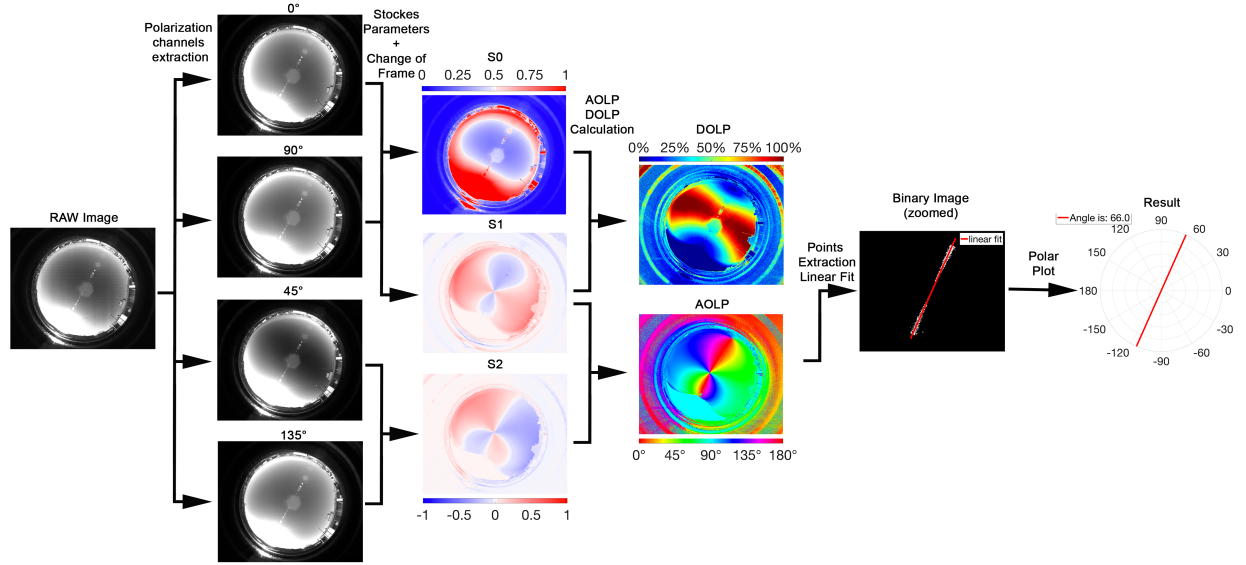


Figure 5: Processing steps to estimate the solar meridian. For illustration purpose, images were taken with a fisheye lens. All Stokes and polarimetric images are already shown in the meridian frame.

4. RESULTS AND ANALYSIS

4.1 Rotary Setup

We have carried out several acquisition campaigns in various conditions. Figure 6 shows an acquisition in clear weather at a speed of +9°/s over a period of 10 s with a data rate of 10 Hz. Figure 6a shows the results of the relative heading given by the algorithm, from the raw images of the FLIR camera. For comparison, the heading

given by the feedback of the rotary stage is also shown. The heading computed by the algorithm is relative because it is based on the solar meridian feature extraction. So, the algorithm returns the angle of the meridian. The first image serves as a reference to find the meridian angle, and all the following images will use the first meridian estimation as a reference. This system allows us to know precisely the heading of the ADRS sensors and therefore to verify the heading given by the camera.

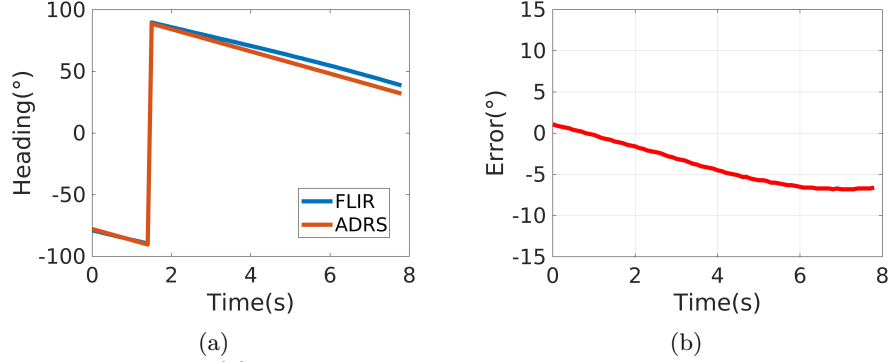


Figure 6: The rotary setup results. (a) Headings computed with the data from the polarimetric camera (FLIR), and the rotary setup (ADRS). (b) Heading error of the image-based navigation system, taking the rotating configuration as a reference. The acquisition campaign was made on 2021/10/06 at 7:56am.

In Figure 6a, we can observe that the curves have the same trend except at the end. Figure 6b shows a maximum error of -6° between the heading estimated using the camera data and the reference given by the rotary stage. This difference can be explained by the levelling of the rotation stage or is due to the growing error in the estimation when the heading get closer to 45° and the matrix can become theoretically non-invertible (Equation (14)). We can also notice that there is a jump for the curves of the cameras around 1.8 s. This is due to the 180° indeterminacy of the sun position. In this case, it might be useful to use the DoLP to remove this ambiguity and thus correct the jump in the heading data. Nevertheless, it is also necessary to evaluate the results in a moving vehicle such as an UGV, which is the aim of the next Section 4.2.

4.2 Embedded setup

We have performed several acquisition campaigns in various conditions (straight line, turns, etc.). This allows us to compare on field data. Figure 7 shows the heading results from both the image-based system (data from the FLIR camera), and that given by the GPS/INS system of the UGV.

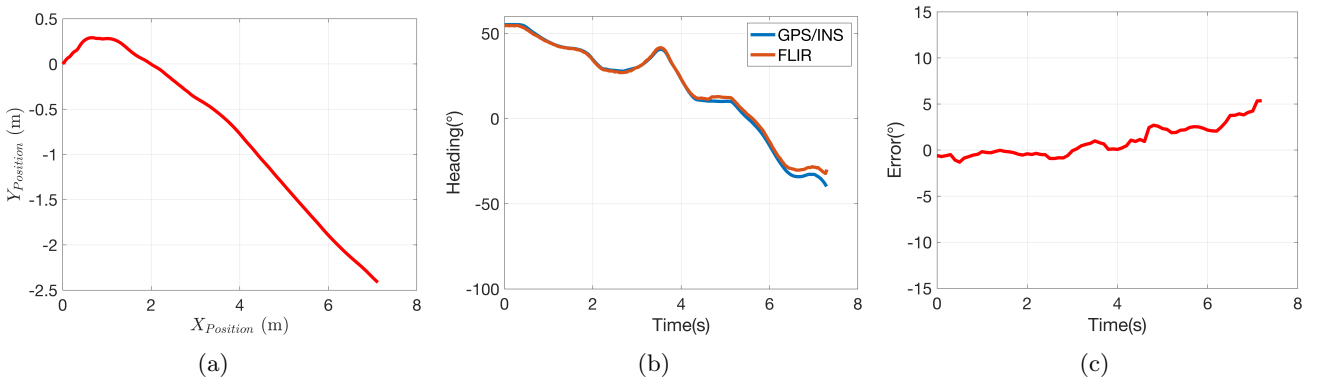


Figure 7: The embedded setup results. (a) UGV trajectory, (b) Headings of the UGV computed from camera data (FLIR), and acquired by GPS/INS. (c) Heading error of the image-based navigation system, taking the GPS/INS data as a reference. The acquisition campaign was on 2021/03/30 at 2:58pm. The datarate of the GPS/INS is of 100 Hz, whereas the datarate of the camera is 10 Hz.

In Figure 7b, we can see that the curves have the same trend, but are not exactly superimposed. Figure 7c shows a maximum error of 5°. This can be partly due to the noise in the image (e.g. saturation), atmospheric conditions (e.g. pollution), levelling of the UGV, or geometrical lens aberrations.

5. CONCLUSION

Based on the polarization pattern of the sky, we developed a navigation solution which combines a polarization camera and an image processing to estimate the heading. The navigation system was tested on both a rotary setup and an embedded setup (UGV). An error of few degrees has been quantified between the estimated and the reference headings.

As a future work, we want to evaluate more deeply the navigation system, in terms of accuracy and robustness. It could include scenarios such as various weather conditions or various optical paths (i.e. lens parameters). Moreover, as the Rayleigh scattering varies with the wavelength, several spectral sensitivities of the acquisition system can be investigated.

Finally, to be able to implement the algorithm in real-time, e.g. for vehicles subject to very high dynamics such as unmanned aerial vehicle, it would be important to reduce the computational cost. One of the solutions would be to spatially subsample the input raw images, and to study the impact of the spatial resolution on the heading estimation accuracy.

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